MILITARY SPECIFICATION

DELAY ELEMENT M53
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for components, loading, assembling and packing of the Delay Element, M53.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-13392	- Primer Percussion
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, Standard Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

FSC 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9246248	- Delay Element Assembly, M53
9282987	- Packing and Marking for Delay Element, M53

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order or precedence. In the event of a conflict between the text of this specification and reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The M53 delay element components and assemblies shall comply with all requirements specified on drawing (dwg.) 9255164, 9246249, 9208228, 9236202, 9208229, and 9246248 and with all the requirements specified in applicable specifications.

3.3 Arming and delay functioning. The delay element shall meet the arming and delay functioning requirements described in 3.3.1 and 3.3.2 respectively.

3.3.1 Arming functioning. The delay element shall fire high order such that the arming pin is ejected when tested as specified in 4.5.2.

3.3.2 Delay functioning. The delay element shall eject the arming pin within a delay time of 1.8 seconds min. to 6.0 seconds max. when tested as specified in 4.5.2 (see 6.4).

3.4 Waterproofness. The delay element shall meet the arming delay functioning requirement (3.3.2) after being subjected to test specified in 4.5.3.

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3.5 Cracks. There shall be no cracks in the body or in the crimped area.

3.6 Foreign material. There shall be no evidence of foreign or explosive material in the primer holder cavity or in the arming pin cavity during assembly. There shall be no evidence of foreign or explosive material in the primer stab area or the arming pin area of the completed item.

3.7 Aluminum tape. The tape shall affect a continuous-unbroken seal over a majority (at least 1/2) of the shoulder surface and not be bent, torn or wrinkled or otherwise deformed.

3.8 Moisture content of explosives. The moisture content before consolidation of the delay mixture, ignitor mixture and pyrotechnic mix shall not exceed 0.50 percent max.

3.9 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, rust, contamination and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items as required per Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Delay Element M53				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9255164) Examination for defects	10		3.2	4.4.2.1
	Pin, Arming (Dwg. 9246249) Examination for defects	10		3.2	4.4.2.2
	Holder, Primer (Dwg. 9208228) Examination for defects	10		3.2	4.4.2.3
	Ring (Dwg. 9236202) Examination for defects	10		3.2	4.4.2.4
	Primer and Holder Assembly (Dwg. 9208229) Examination for defects	10		3.2	4.4.2.5
	Delay Element Assembly, M53 (Prior to assembly of arming pin, primer, and holder assembly) (Dwg. 9245248) Examination for defects	10		3.2/3.6/ 3.7	4.4.2.6

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Delay Element M53				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Element Assembly, M53 (Dwg. 9246248)				
	Examination for defects	10		3.2	4.4.2.7
	Arming and delay functioning test (70°F)	100		3.3.1/ 3.3.2	4.5.2
	Waterproofness test	50		3.4	4.5.3
	Arming pin pull test	25		3.2	4.5.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of delay elements shall contain:

a. Delay metal parts from one lot interfix from one manufacturer.
b. M54 primers from one lot interfix from one manufacturer.
c. Explosives that come from one batch or blending of batches which result in homogeneity of all explosives used in the loaded lot.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9255164
4.4.2.1	Body				NEXT HIGHER ASSEMBLY 9246248
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Diameter of arming pin hole, min.		0.40%	3.2	Gage
103	Diameter of pyrotechnic hole		0.40%	3.2	Gage
104	Diameter of delay hole		0.40%	3.2	Gage
105	Diameter of primer hole		0.40%	3.2	Gage
106	Overall length		0.40%	3.2	Gage
107	Depth of arming pin hole, min.		0.40%	3.2	Gage
108	Length of pyrotechnic hole		0.40%	3.2	Gage
109	Depth of primer hole, min.		0.40%	3.2	Gage
110	True position of delay material hole		0.40%	3.2	Gage
111	Perpendicularity of base of arming pin hole with outside diameter		0.40%	3.2	Gage
112	True position of pyrotechnic hole		0.40%	3.2	Gage
113	True position of delay hole		0.40%	3.2	Gage
114	True position of primer hole		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Pin, Arming	1 OF 1	9246249
			NEXT HIGHER ASSEMBLY 9246248

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Distance to head		0.40%	3.2	Gage
103	Diameter of piston		0.40%	3.2	Gage
104	Perpendicularity of piston		0.40%	3.2	Gage
105	True position of head		0.40%	3.2	Gage
106	Height of piston, min.		0.40%	3.2	Gage
107	Diameter of shaft, min.		0.40%	3.2	Gage
108	Diameter of head, max.		0.40%	3.2	Gage
109	Finish improper		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9208228
4.4.2.3	Holder, Primer				NEXT HIGHER ASSEMBLY 9208229
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter, min.		0.40%	3.2	Gage
102	Hole diameter, min.		0.40%	3.2	Gage
103	Outside diameter, max.		0.40%	3.2	Gage
104	Overall length		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9236202
4.4.2.4	Ring				NEXT HIGHER ASSEMBLY 9246248
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Outside diameter, max.		0.40%	3.2	Gage
103	Inside diameter, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9208229
4.4.2.5	Primer and Holder Assembly				NEXT HIGHER ASSEMBLY 9246248
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Inverted primer, percussion		0.40%	3.2	Visual
103	Primer hole missing or plugged		0.40%	3.2	Visual
104	Primer below flush with surface of holder		0.40%	3.2	Gage
105	Outside diameter, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual
4.4.2.6	Explosive Mixes (				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Explosive Mixes (delay, ignitor, and pyrotechnic) Moisture Test		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Moisture content excessive		1/	3.8	4.5.5
Minor	None defined				

NOTES: 1/ 0.4 to 0.7 gram shall be selected from each batch (see 4.3.1.c). The batch represented by the sample shall be rejected if the sample fails to meet the requirement.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Delay Element Assembly, M53, Prior to Assembling of Arming Pin, Primer, and Holder Assembly	1 OF 1	9246248
			NEXT HIGHER ASSEMBLY 9246246

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Missing pyrotechnic mix		0.40%	3.2	Visual
102	Insufficient coverage of sealing surface by aluminum tape		0.40%	3.7	Visual
103	Evidence of foreign or explosive material in primer holder cavity or arming pin cavity		0.40%	3.6	Visual
104	Ignitor missing from ring		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9246248
4.4.2.8	Delay Element, M53				NEXT HIGHER ASSEMBLY 9246246
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Arming pin missing		100%	3.2	Visual
2	Any of three arming pin stakes missing or incomplete		100%	3.2	Visual
Major					
101	Concentricity of arming pin with body		0.40%	3.2	Gage
102	Overall length, max.		0.40%	3.2	Gage
103	Primer missing		0.40%	3.2	Visual
104	Presence of cracks in the crimp		0.40%	3.5	Visual
105	Crimp missing or incomplete		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9246248
4.4.2.8.1	Delay Element, M53 (Testing)				NEXT HIGHER ASSEMBLY 9246246
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Delay functioning, short time at 160°F		1/	3.3.2	4.5.2
2	Delay functioning, short time at 70°F		1/	3.3.2	4.5.2
3	Delay functioning, short time at -65°F		1/	3.3.2	4.5.2
4	Arming pin pull test, below minimum load	500	2/	3.2	4.5.4
Major					
101	Arming functioning, non-functioning at 160°F		0.25%	3.3.1	4.5.2
102	Arming functioning, non-functioning at 70°F		0.25%	3.3.1	4.5.2
103	Arming functioning, non-functioning at -65°F		0.25%	3.3.1	4.5.2
104	Delay functioning, above max. time at 160°F		0.25%	3.3.2	4.5.2
105	Delay functioning, above max. time at 70°F		0.25%	3.3.2	4.5.2
106	Delay functioning, above max. time at -65°F		0.25%	3.3.2	4.5.2
107	Arming pin pull test, failure to move above maximum load	500	2/	3.2	4.5.4
108	Waterproofness test	60	3/	3.4	4.5.3

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-14951A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.8.1	Delay Element, M53 (Testing)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ Sampling plan and units shall be the same as the Arming Functioning Test; except the Criticals shall have no defects permitted. 2/ Compute pull load mean ($\bar{x}$) and standard deviation (unbiased) (s). Rejection criteria is as follows: a. $\bar{x} - 5s > 22$ b. $\bar{x} + 2.5s < 80$ c. $\bar{x} \geq 40$ d. Any individual (xi) less than 30. e. Any two (2) individuals (xi) greater than 80. 3/ The lot shall be rejected if one or more samples fails to meet the requirements.				

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.9	Packing and Marking for Delay Element, M53		1 OF 1		9282987
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Set-up box damaged exposing contents		0.40%	3.2	Visual
102	Set-up box improperly marked		0.40%	3.2	Visual
103	Waterproofness of barrier bag destroyed by rip, tear, puncture, or separation of heat seal		0.40%	3.2	Visual
104	Barrier bag improperly marked		0.40%	3.2	Visual
105	Strapping missing or loose		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and thal all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Arming and delay time functioning. The sample for each temperature phase shall be conditioned in the fixture at 160 ± 5 degrees Fehrenheit(F), 70 ± 5 degrees F and -65 ± 5 degrees F respectively for a minimum of 24 hours. The delay element shall be assembled to a fixture and fired within 30 seconds of conditioning by dropping a two ounce max. ball in the test fixture from a height of eight inches, max. For the delay functioning test, equipment shall be capable of measuring from the ignition of the primer to the flash of the pyrotechnic mix. Any item failing to meet the requirements shall be classified defective.

4.5.3 Waterproofness test. The samples shall be placed into a container and submerged in a 4" min. of water at 70°F ± 10°F for 48 hours. Within one hour after removal from the water, the element shall be subjected to the delay functioning test at ambient temperature.

4.5.4 Pull test of arming pin. The delay element shall be held in an appropriate fixture and an axial load shall be

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applied until the arming pin pulls out from the delay element. The load shall be recorded. All samples subjected to this test shall be scrapped (destructive test). If the requirements of 4.4.2.8.1 are not met, the lot shall be rejected.

4.5.5 Moisture content of explosives. A known value of between 0.4 to .07 gram of the sample shall be transferred to a tared weighing dish approximately 55 millimeter (mm) in diameter and 30mm in height. Place the dish and contents in an oven maintained at 55 to 65 degrees Centigrade (C.) or in a desiccator containing anhydrous calcium chloride until constant weight is obtained. Calculate the loss in weight as the percentage moisture.

4.5.6 Check test for deterioration of primer. If the total time between original acceptance of the primer lot and the assembly of that lot into delay elements exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must pass the check test for deterioration specified in MIL-P-13392 (see 6.5).

5. PACKAGING.

5.1 Packaging. The delay element shall be packaged in accordance with drawing 9282987.

5.2 Packing. The delay elements shall be packed in accordance with drawing 9282987.

5.3 Marking. Marking shall be in accordance with drawing 9282987.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the M567 fuze for the 81mm Cartridge, HE, M374 Series and WP, M375 Series.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDR-QAT-I, Dover, NJ 07801.

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6.4 If any element fires before the 1.8 seconds minimum, the element shall be classified as a short time defect. If any element is fired after six seconds or does not fire at all, the element shall be classified as a long time defect.

6.5 Check test for deterioration of primer. This test shall be perfomed as per MIL-P-13392 by the contractor on primers selected by the Government inspector at the facility assembling the primer into the delay element. The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or the contractor assembling the primers into the delay element and shall not constitute rejection except when deterioration has occurred as a direct result of carelessness in handling, storage, etc. committed while the primer lot was under the jurisdiction of either contractor.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1390-A348)

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MIL-D-14951A Delay Element M53 Loading, Assembling and Packing

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